

```
[4]: marks = [75, 82, 90, 68, 88]
marks.append(95)
marks.insert(2, 80)
marks.remove(68)
marks.sort()
print("Updated Marks List:", marks)
```

Updated Marks List: [75, 80, 82, 88, 90, 95]

```
[3]: cities = ("Delhi", "Mumbai", "Chennai", "Bangalore", "Hyderabad", "Kolkata")
print("First City:", cities[0])
print("Last City:", cities[-1])
print("Total Elements:", len(cities))
if "Delhi" in cities:
    print("Delhi exists")
else:
    print("Delhi does not exist")
```

First City: Delhi
Last City: Kolkata
Total Elements: 6
Delhi exists

```
[5]: A = {1, 2, 3, 4, 5}
B = {4, 5, 6, 7}
print("Union:", A.union(B))
print("Intersection:", A.intersection(B))
print("Difference:", A.difference(B))
print("Symmetric Difference:", A.symmetric_difference(B))
```

Union: {1, 2, 3, 4, 5, 6, 7}
Intersection: {4, 5}
Difference: {1, 2, 3}
Symmetric Difference: {1, 2, 3, 6, 7}

```
[7]: student = {  
    "name": "Manisha",  
    "roll": 101,  
    "branch": "CSE",  
    "marks": 95  
}  
student["marks"] = 95  
print("Keys:")  
for key in student.keys():  
    print(key)  
print("Values:")  
for value in student.values():  
    print(value)
```

```
Keys:  
name  
roll  
branch  
marks  
Values:  
Manisha  
101  
CSE  
95
```

```
[8]: items = ["Bag", "Book", "Pen"]  
prices = [500, 200, 20]  
total_bill = sum(prices)  
print("Total Bill:", total_bill)  
print("Highest Price:", max(prices))  
print("Lowest Price:", min(prices))
```

```
Total Bill: 720  
Highest Price: 500  
Lowest Price: 20
```

```
[9]: def addition(a, b):  
      return a + b  
      def subtraction(a, b):  
          return a - b  
      def multiplication(a, b):  
          return a * b  
      def division(a, b):  
          return a / b  
      a = int(input("Enter first number: "))  
      b = int(input("Enter second number: "))  
      print("Addition:", addition(a, b))  
      print("Subtraction:", subtraction(a, b))  
      print("Multiplication:", multiplication(a, b))  
      print("Division:", division(a, b))
```

```
Enter first number: 2  
Enter second number: 5  
Addition: 7  
Subtraction: -3  
Multiplication: 10  
Division: 0.4
```

```
[10]: def check_even_odd(n):  
       if n % 2 == 0:  
           return "Even"  
       else:  
           return "Odd"  
       num = int(input("Enter number: "))  
       print(check_even_odd(num))
```

```
Enter number: 3  
Odd
```

```
[11]: def largest(a, b, c):  
    if a >= b and a >= c:  
        return a  
    elif b >= a and b >= c:  
        return b  
    else:  
        return c  
print("Largest:", largest(10, 25, 15))
```

Largest: 25

```
[12]: def factorial(n):  
    fact = 1  
    for i in range(1, n + 1):  
        fact *= i  
    return fact  
num = int(input("Enter number: "))  
print("Factorial:", factorial(num))
```

Enter number: 4

Factorial: 24

```
[13]: def is_prime(n):
        if n <= 1:
            return False
        for i in range(2, n):
            if n % i == 0:
                return False
        return True

num = int(input("Enter number: "))
if is_prime(num):
    print("Prime Number")
else:
    print("Not Prime Number")
```

Enter number: 5
Prime Number

```
[14]: students = []

def add_student(name):
    students.append(name)
def remove_student(name):
    if name in students:
        students.remove(name)
def display_students():
    print("Students:", students)
add_student("Manisha")
add_student("Bhavana")
display_students()
remove_student("Bhavana")
display_students()
```

Students: ['Manisha', 'Bhavana']
Students: ['Manisha']

```
[16]: employees = {}  
def add_employee(name, salary):  
    employees[name] = salary  
def update_salary(name, salary):  
    if name in employees:  
        employees[name] = salary  
def search_employee(name):  
    if name in employees:  
        print(name, "Salary:", employees[name])  
    else:  
        print("Employee not found")  
add_employee("Ram", 30000)  
update_salary("Ram", 35000)  
search_employee("Ram")
```

Ram Salary: 35000

```
[17]: numbers = []  
for i in range(10):  
    num = int(input("Enter number: "))  
    numbers.append(num)  
unique_numbers = set(numbers)  
print("Unique Numbers:", unique_numbers)  
print("Count of Unique Numbers:", len(unique_numbers))
```

```
Enter number: 3  
Enter number: 4  
Enter number: 5  
Enter number: 6  
Enter number: 7  
Enter number: 8  
Enter number: 9  
Enter number: 10  
Enter number: 5  
Enter number: 6  
Unique Numbers: {3, 4, 5, 6, 7, 8, 9, 10}  
Count of Unique Numbers: 8
```

```
[18]: ages = (12, 18, 25, 16, 30)
def check_voting(age):
    if age >= 18:
        return "Eligible"
    else:
        return "Not Eligible"
for age in ages:
    print(age, "-", check_voting(age))
```

```
12 - Not Eligible
18 - Eligible
25 - Eligible
16 - Not Eligible
30 - Eligible
```

```
[19]: sentence = input("Enter sentence: ")
words = sentence.split()
frequency = {}
for word in words:
    if word in frequency:
        frequency[word] += 1
    else:
        frequency[word] = 1
print(frequency)
```

```
Enter sentence: python is easy subject
{'python': 1, 'is': 1, 'easy': 1, 'subject': 1}
```

```
[20]: books = []
def add_book(book):
    books.append(book)
def remove_book(book):
    if book in books:
        books.remove(book)
def search_book(book):
    if book in books:
        print("Book Found")
    else:
        print("Book Not Found")
def display_books():
    print("Books:", books)
add_book("Python")
add_book("Java")
display_books()
search_book("Python")
remove_book("Java")
display_books()
```

Books: ['Python', 'Java']

Book Found

Books: ['Python']


```
[21]: marks = []
for i in range(5):
    m = int(input("Enter marks: "))
    marks.append(m)
def average_marks(marks):
    return sum(marks) / len(marks)
def highest_marks(marks):
    return max(marks)
def lowest_marks(marks):
    return min(marks)
def passed_students(marks):
    count = 0
    for mark in marks:
        if mark > 40:
            count += 1
    return count
print("Average:", average_marks(marks))
print("Highest:", highest_marks(marks))
print("Lowest:", lowest_marks(marks))
print("Passed Students:", passed_students(marks))
```

```
Enter marks: 95
Enter marks: 96
Enter marks: 91
Enter marks: 98
Enter marks: 99
Average: 95.8
Highest: 99
Lowest: 91
Passed Students: 5
```

```
[23]: def check_password(password):
    has_digit = False
    has_upper = False
    has_lower = False
    for ch in password:
        if ch.isdigit():
            has_digit = True
        elif ch.isupper():
            has_upper = True
        elif ch.islower():
            has_lower = True
    if len(password) >= 8 and has_digit and has_upper and has_lower:
        return "Strong Password"
    else:
        return "Weak Password"
pwd = input("Enter password: ")

print(check_password(pwd))
```

Enter password: 78947

Weak Password

```
[24]: contacts = {}  
def add_contact(name, number):  
    contacts[name] = number  
def search_contact(name):  
    if name in contacts:  
        print(name, ":", contacts[name])  
    else:  
        print("Contact not found")  
def delete_contact(name):  
    if name in contacts:  
        del contacts[name]  
def display_contacts():  
    print(contacts)  
add_contact("Manisha", "9876543210")  
add_contact("Bhavana", "9123456780")  
display_contacts()  
search_contact("Manisha")  
delete_contact("Bhavana")  
display_contacts()  
  
{'Manisha': '9876543210', 'Bhavana': '9123456780'}  
Manisha : 9876543210  
{'Manisha': '9876543210'}
```

```
[28]: balance = 1000

def deposit(amount):
    global balance
    balance += amount
    print("Deposited:", amount)
def withdraw(amount):
    global balance
    if amount <= balance:
        balance -= amount
        print("Withdrawn:", amount)
    else:
        print("Insufficient Balance")
def check_balance():
    print("Current Balance:", balance)
deposit(500)
withdraw(300)
check_balance()
```

Deposited: 500

Withdrawn: 300

Current Balance: 1200

```
[29]: numbers = [10, 20, 30, 40, 50]
maximum = numbers[0]
minimum = numbers[0]
for num in numbers:
    if num > maximum:
        maximum = num
    if num < minimum:
        minimum = num
print("Maximum:", maximum)
print("Minimum:", minimum)
reversed_list = []
for i in range(len(numbers) - 1, -1, -1):
    reversed_list.append(numbers[i])
print("Reversed List:", reversed_list)
```

Maximum: 50

Minimum: 10

Reversed List: [50, 40, 30, 20, 10]